Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 07)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A mining company is assessing a copper deposit beneath farmland.

0 1 . 1 Describe what each of the following survey techniques detects.

[3 marks]

Gravimetry .....

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Magnetometry .....

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Seismic surveys .....

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0 1 . 2 Explain why trial drilling is normally carried out only after the remote sensing surveys.

[2 marks]

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0 1 . 3 Explain how a rise in the world price of copper can turn part of the **resource base** into a **reserve**.

[2 marks]

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0 1 . 4 Explain **four** environmental impacts of extracting the copper by opencast mining.

[4 marks]

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0 1 . 5 Describe how the site could be restored after closure, and explain why restoration is planned before mining begins.

[4 marks]

15

0 2 . 1 Explain why current global energy use is described as unsustainable.

[3 marks]

0 2 . 2 Describe the environmental impacts of exploiting fossil fuels that occur **before** the fuel is used.

[4 marks]

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0 2 . 3 Explain why an 'energy gap' could occur if society waits until fossil fuels become expensive before developing alternatives.

[3 marks]

10

0 3 . 1 Explain why compressed air energy storage with heat recovery can achieve over 90% efficiency.

[2 marks]

0 3 . 2 Explain why large-volume hot water stores lose heat more slowly per unit of stored energy than small ones.

[2 marks]

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0 3 . 3 Explain why molten salt rather than water is used to store heat from CSP plants.

[2 marks]

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0 3 . 4 Explain how high thermal mass building materials reduce energy use.

[2 marks]

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0 3 . 5 Describe **one** other energy storage technology.

[2 marks]

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0 4 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☒

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☒

D To measure the concentration of a pollutant at the point of release ☐

0 4 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 4 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

5

0 5 A coal-mine spoil heap on a hillside has been abandoned. The spoil contains shale with some iron pyrites and residual coal.

0 5 . 1 Explain **three** problems this spoil heap could cause and describe how each could be managed.

[6 marks]

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0 5 . 2 Explain why deep layers of soil are applied when a spoil heap containing toxic materials is restored.

[2 marks]

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0 5 . 3 Explain why an old steel-works spoil heap may be valuable for wildlife conservation.

[2 marks]

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0 6 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 6 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 7 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 7 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 7 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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0 8 Figure 1 is a flow diagram of the main stages in a sewage treatment works using activated sludge treatment.

Figure 1

Stage	Process
Pre-treatment	Screens and grit traps
Primary treatment	Primary sedimentation
Secondary treatment	Aeration tank then secondary sedimentation
Tertiary treatment	Phosphate stripping / micro-strainers / UV
Sludge treatment	Anaerobic digestion

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0 8 . 1 Explain the purpose of screens and grit traps.

[2 marks]

0 8 . 2 Explain why primary sedimentation is described as the most important process.

[2 marks]

0 8 . 3 Describe what happens in the aeration tank and explain why the sludge from the secondary sedimentation tank is returned to it.

[3 marks]

0 8 . 4 Explain why tertiary treatment may be needed before effluent is discharged, and describe **one** tertiary process.

[3 marks]

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0 9 . 1 Explain how each of the following human activities increases soil erosion.

[4 marks]

Overgrazing .....

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Reducing soil biota .....

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0 9 . 2 Explain why landslides become more likely on steep slopes some years after deforestation.

[2 marks]

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0 9 . 3 Describe **four** environmental impacts of accelerated soil erosion away from the field where it occurs.

[4 marks]

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1 0 Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O_2)	2
Triatomic oxygen / ozone (O_3)	3

1 0 . 1 Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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1 0 . 2 Give **two** other ways in which the atmosphere supports life.

[2 marks]

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1 0 . 3 Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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1 0 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

1 1 . 2 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS